(a) $f$ (b) $\frac{f}{2}$

(c) $\frac{f}{f+2}$ (d) $\frac{f+2}{f}$

**9.** If the temperature and pressure of a gas is doubled the mean free path of the gas molecules

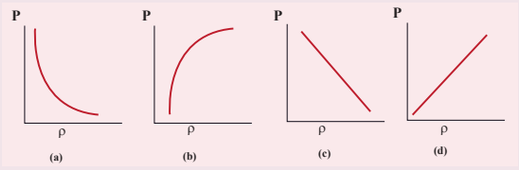
(a) remains same

(b) doubled

(c) tripled

(d) quadrapoled

**10.** Which of the following shows the correct relationship between the pressure and density of an ideal gas at constant temperature?




**11.** A sample of gas consists of $\mu_1$ moles of monoatomic molecules, $\mu_2$ moles of diatomic molecules and $\mu_3$ moles of linear triatomic molecules. The gas is kept at high temperature. What is the total number of degrees of freedom?

(a) $[3\mu_1 + 7(\mu_2 + \mu_3)]\, N_A$

(b) $[3\mu_1 + 7\,\mu_2 + 6\mu_3]\, N_A$

(c) $[7\mu_1 + 3(\mu_2 + \mu_3)]\, N_A$

(d) $[3\mu_1 + 6(\mu_2 + \mu_3)]\, N_A$

**12.** If $s_P$ and $s_V$ denote the specific heats of nitrogen gas per unit mass at constant pressure and constant volume respectively, then (JEE 2007)

(a) $s_P - s_V = 28R$ (b) $s_P - s_V = R/28$

(c) $s_P - s_V = R/14$ (d) $s_P - s_V = R$

**13.** Which of the following gases will have least rms speed at a given temperature?

(a) Hydrogen

(b) Nitrogen

(c) Oxygen

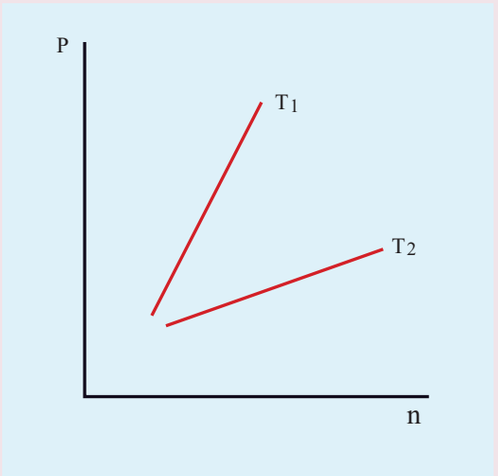
(d) Carbon dioxide

**14.** For a given gas molecule at a fixed temperature, the area under the Maxwell-Boltzmann distribution curve is equal to

(a) $\frac{PV}{kT}$ (b) $\frac{kT}{PV}$

(c) $\frac{P}{NkT}$ (d) $PV$

**15.** The following graph represents the pressure versus number density for ideal gas at two different temperatures $T_1$ and $T_2$. The graph implies




(a) $T_1 = T_2$

(b) $T_1 > T_2$

(c) $T_1 < T_2$

d) Cannot be determined

**Answers:**

| | | | |
|---|---|---|---|
| 1) a | 2) c | 3) b | 4) a |
| 5) a | 6) b | 7) c | 8) d |
| 9) a | 10) d | 11) a | 12) b |
| 13) d | 14) a | 15) b | |



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